

Sukh Sandhu Layman Auditor Guide

Plain-English instructions for reviewing the Udacity AI Capstone defense package

Reader assumption

This guide assumes the auditor has no prior knowledge of Udacity, artificial intelligence, GitHub, Python, notebooks, or how project files were created and run.

1. What this package is

This is a final defense package for a Udacity AI Capstone project. A capstone is a final project that brings together skills learned across a program. Project 8 is a presentation and defense of Project 7, which designed an integrated AI system for healthcare operations triage.

The project is not a medical product. It is a learning and portfolio artifact showing how different AI methods can be combined into a responsible workflow.

2. Basic terms for a complete beginner

Term	Plain-English meaning
Udacity	The online learning platform that provides the course, instructions, rubric, and submission portal.
Rubric	The marking checklist. It says what must be present for review.
AI	Software that uses data or rules to support predictions, classifications, summaries, or decisions.
GitHub	A public website used to store code, file history, and project repositories.
Repository	A project folder on GitHub containing code, documentation, and version history.
Branch	A separate line of work inside a repository.
Commit	A saved checkpoint showing progress over time.
Python	The programming language used to run scripts and notebooks.
Notebook	An interactive document mixing code, explanations, tables, and outputs.
requirements.txt	A list of software packages needed to rerun the project.
JSON	A structured text file that stores results.
Zip file	A compressed package containing the final files for upload.

3. What the project does in plain English

- 1 A healthcare operations case comes in.
- 2 The system checks whether the case has data-quality problems.
- 3 The system considers whether a machine-learning model is confident.
- 4 If visual material is relevant, the system considers whether an image model is confident or uncertain.
- 5 A generative AI quality signal flags whether the evidence is in-distribution or an outlier.
- 6 The agentic routing layer combines the signals and chooses a review pathway.
- 7 If a case is patient-facing, the system requires human approval.

4. How the project was done step by step

Step	What was done	Why it matters
1	Selected healthcare operations as the industry context.	Gives the project a real-world setting.
2	Defined the problem as safe case routing when evidence quality varies.	Makes the system useful and realistic.
3	Mapped prior capstone projects to system components.	Shows synthesis across the program.
4	Created an integrated routing script for Project 7.	Turns the design into observable behavior.
5	Ran four test scenarios and saved outputs to JSON.	Creates evidence an auditor can inspect.
6	Prepared a reflective synthesis paper for Project 7.	Explains design, ethics, and professional relevance.
7	Prepared a defense deck and speaker notes for Project 8.	Supports the mentor presentation.
8	Published repositories on GitHub.	Provides public evidence and version history.
9	Packaged required files into zip archives.	Creates upload-ready Udacity files.

5. What each main file is for

File	Purpose
Sukh_Sandhu_Detailed_Professional_Industry_Defense_Deck.pptx	Detailed PowerPoint presentation with speaker notes.
Sukh_Sandhu_Defense_Discussion_and_QA_Guide.pdf	Defense speaking guide and mentor question preparation.
Sukh_Sandhu_Layman_Auditor_Guide.pdf	Plain-English audit guide.
Professional_Industry_Defense_Deck.pptx	Original concise Project 8 deck.
Professional_Industry_Defense_Speaker_Notes.pdf	Original concise speaker notes.
README.md	Short explanation of the project folder and repository.
requirements.txt	Software dependencies.
repository_evidence.txt	GitHub repository and branch evidence.
integrated_scenario_results.json	Project 7 output showing four scenario decisions.

6. How to review without coding

- 1 Open the Project 8 zip file.
- 2 Confirm the deck, notes, guides, README, requirements, and repository evidence are present.
- 3 Open the detailed PowerPoint deck and check that it covers context, system overview, integration, technical decisions, evaluation, ethics, limitations, and closing readiness.
- 4 Open the discussion guide and check that it contains timing, slide-by-slide notes, and mentor Q&A preparation.
- 5 Open the auditor guide and check that it explains the project in plain English.
- 6 Open the public GitHub repository link and confirm that the code and supporting files are visible.
- 7 Compare the contents to the Udacity rubric.

7. How to run Project 7 code if Python is available

Action	Instruction
Install dependencies	Open a terminal in the Project 7 folder and run: <code>pip install -r requirements.txt</code>
Run the system	Run: <code>python integrated_health_ai_system.py</code>
Check output	Open <code>outputs/integrated_scenario_results.json</code>
Expected result	Four scenarios should appear: standard operations queue, data quality review, human approval required, and expedited review.

8. What the four scenarios mean

Scenario	Risk score	Decision	Plain-English explanation
SYN-001	0.12	Standard operations queue	Evidence quality and confidence are acceptable.
SYN-002	0.62	Data quality review	Data quality and uncertainty require review.
SYN-003	0.34	Human approval required	The safety boundary applies because the case is patient-facing.
SYN-004	0.98	Expedited review	Multiple caution signals produce very high risk.

9. How to compare against the rubric

Rubric area	What to look for
Presentation structure	Clear sequence and 15-minute plan.
Industry problem	Healthcare operations problem is explained before technical details.
System explanation	Major components and interactions are described.
Prior project integration	At least three prior capstone projects are named and connected.
Technical decisions	Tradeoffs, limitations, and alternatives are discussed.
Ethics	Risks are connected to safeguards.
Evaluation	Scenario outputs and failure cases are discussed.
Professional readiness	Presenter can explain the project to technical and non-technical audiences.

10. Auditor red flags

- Presenter claims clinical validation.
- Presenter cannot explain human approval boundaries.
- Deck does not connect prior projects to the final system.
- Repository link is inaccessible.
- Scenario output file is missing.
- Ethics discussion is generic rather than tied to this system.

11. Final auditor conclusion

A lay auditor should conclude that the package is review-ready when the files are present, the presentation explains the required sections, the scenario evidence matches the deck, the GitHub repository is public, and the presenter can explain limitations honestly. The live defense itself still depends on the actual spoken presentation and answers given during mentor questioning.